

Neurocase

The Neural Basis of Cognition

ISSN: (Print) (Online) Journal homepage: <https://www.tandfonline.com/loi/nncs20>

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To cite this article: Katherine Reiter & Kasia Gustaw Rothenberg (2020): Neuropsychological presentation of colpocephaly and porencephaly with symptom onset in adulthood, *Neurocase*, DOI: [10.1080/13554794.2020.1841798](https://doi.org/10.1080/13554794.2020.1841798)

To link to this article: <https://doi.org/10.1080/13554794.2020.1841798>



Published online: 02 Nov 2020.



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Neuropsychological presentation of colpocephaly and porencephaly with symptom onset in adulthood

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ABSTRACT

Colpocephaly is a form of congenital ventriculomegaly while porencephaly describes any full-thickness defect within the brain which usually presents as a cystic structure. Postulated aetiologies include intrauterine/perinatal injuries, genetic disorders, and morphogenesis error. Colpocephaly and porencephaly is typically diagnosed in infancy while diagnosis in adulthood is exceptionally rare. We report a case of co-existence of colpocephaly with porencephaly diagnosed incidentally in a 54-year-old male presenting with subtle cognitive and neurologic abnormalities. Neuropsychological assessment revealed weaknesses in executive functions, processing speed, and language. To our knowledge, this is the only reported case of dual incidental findings of porencephaly and colpocephaly in an adult.

ARTICLE HISTORY

Received 18 May 2020
Accepted 18 October 2020

KEYWORDS

Porencephaly; colpocephaly;
neuropsychology; adult;
neuroimaging

Introduction

Colpocephaly is a rare form of congenital ventriculomegaly. First described by Benda (1941) as rare congenital brain malformation in which the occipital horns are disproportionately larger than the anterior horns of the lateral ventricles. The term was coined in the 1940s based on the Greek term “kolpos” meaning hollow. Colpocephaly was further described as a persistence of fetal cerebral ventricles into postnatal life (Puvabanditsin et al., 2006). Since this first description in the 1940s, multiple cases have been reported (Cerullo et al., 2000; Patnaik et al., 2012; Saldanha et al., 2017), rarely, however, in adulthood (Esenwa & Leaf, 2013; Nasrat & Seraji-Bozoergzad, 2015).

Porencephaly is a rare congenital disorder of the central nervous system involving a cyst or a cavity within the cerebral parenchyma filled with cerebrospinal fluid (Berg et al., 1983). The term “porencephaly” describes any full-thickness defect within the brain cortex, irrespective of etiology (Stewart et al., 1978). Porencephalic pathology is generally divided into two categories based on etiology and time of origin during the pregnancy. Type I porencephaly (referred to in the literature as false porencephaly, encephaloclastic porencephaly, or encephaloclastic schizencephaly) is an acquired condition thought to occur in the third trimester due to a destructive process such as periventricular hemorrhage or central nervous system (CNS) infection. The area of destroyed cerebral parenchyma is replaced with CSF. Type II porencephaly, referred to as true porencephaly, schizencephalic porencephaly, or developmental schizencephaly, is a congenital condition thought to be due to a defect in neuronal migration, usually occurring in the second trimester (Berg et al., 1983).

The etiologies of colpocephaly as well as porencephaly are diverse. There is a strong genetic link to porencephaly, including mutations on COL4A1 and COL4A2 (Verbeek et al., 2012; Yoneda et al., 2012, 2013). Carriers for these mutations often show comorbid cerebrovascular pathology, including both ischemic and

hemorrhagic events (Van der Knaap et al., 2006; Gunda et al. 2014). Porencephaly may present as various symptoms. Pathology was reported to be comorbid with epilepsy especially in the course of hippocampal sclerosis (Burneo et al., 2003; Ho et al., 1998). Multiple comorbid psychiatric syndromes and symptoms are reported, including psychosis and anger outbursts (Chamard et al., 2016; Douzenis et al., 2010; Hussain et al., 2015; Singh et al., 2018). The initial signs and symptoms of porencephaly depend on the location of the defect and include convulsions (Berg et al., 1983; Qureshi et al., 2018; Thafar et al., 2017), varying degrees of developmental delay (Berg et al., 1983; Devi et al., 2002; Singh et al., 2018), visual and sensory defects (Berg et al., 1983; Danckert et al., 2004; Ryzenman et al., 2007; Yang et al., 2012), focal motor deficits (Berg et al., 1983; Blackman et al., 1991; Sherlock et al., 2008), hydrocephalus (Suarez et al., 1984; Tomà et al., 1990; Van der Knaap et al., 2006), and vascular syndromes (Murray et al., 2014; Van der Knaap et al., 2006; Verbeek et al., 2012).

Colpocephaly is often associated with partial or complete agenesis of the corpus callosum (Bartolome et al., 2016; Nasrat & Seraji-Bozoergzad, 2015) and may present as a seizure disorder (Oommen et al., 2017; Wunderlich et al., 1996). When identified incidentally, colpocephaly may be misdiagnosed as hydrocephalus.

Both porencephaly and colpocephaly are typically diagnosed in infancy due to symptom severity. Specific symptoms include motor, sensory, cognitive difficulties, developmental delay, seizures, and/or behavioral concerns (Herskowitz et al., 1985; Landman et al., 1989). Discovery of colpocephaly and porencephaly in adulthood is exceedingly rare (Esenwa & Leaf, 2013). Recently Parker et al. (2019) reported a case of colpocephaly in a neurologically normal adult, incidentally discovered in the emergency department. Cognitive findings are limited in adult diagnoses of porencephaly/colpocephaly but some literature has shown preserved verbal abilities with normal IQ (Hussain et al., 2015), and weaknesses in visuospatial

abilities (Danckert et al., 2004), , attention, and executive functioning (Chamard et al., 2016). Of note, these case studies include multiple medical comorbidities (e.g., seizures) and genetic underpinnings that may limit the generalizability of these findings (See Table 1 for previously reported cases of colpocephaly and porencephaly diagnosed in adulthood).

Materials and methods

History of present illness

This 54-year-old man was referred for comprehensive neuropsychiatric/neuropsychological assessment of subtle cognitive and motor neurological symptoms as well as new-onset impulse control difficulties.

Specifically, he has an approximately 3-year history of motor changes, described as a right leg foot drag, a right-hand tremor that is noticeable at rest, and weakness in both arms, such that he had to reduce his weight lifting while exercising by 10 pounds on both sides. He has muscle stiffness when rising from a seated position. He experiences imbalance and near-falls every 1–2 weeks.

He has verbal outbursts that occur approximately 1–2 times per month. His wife noted some “odd” behaviors that began over the last year, including mumbling and smiling to himself. She described that he sometimes laughs when no one is around. Per his wife, he denies that he is responding to internal stimuli during these times. He reported no current hallucinations or delusions. Paranoia was difficult to assess due to the patient’s guarded communication style. However, his family provided some examples that were concerning for paranoia, including that the patient requests all cell phones to be completely turned off during doctor’s appointments due to concern that others might hear the conversation. He has a longstanding tendency toward ritualistic behaviors, such as needing items placed in specific locations, turning the light switch on and off several times, etc. He reported mild distress and agitation when these activities are not completed. He denied symptoms of frank anxiety or depression. He is working with a mental health provider and per the patient’s wife, this provider has reportedly raised concern for multiple diagnoses,

including Attention-Deficit/Hyperactivity Disorder, Asperger syndrome, Obsessive Compulsive Disorder, Schizoaffective Disorder, and Borderline Personality Disorder. Records from these visits were unavailable to review. The patient reported that he feels Asperger Syndrome is consistent with his experiences, though social developmental was reported to be normal and he is able to maintain current relationships. Sports was described as a possible perseverative interest, though it is not clear that this met clinical significance.

Medical history

Hyperlipidemia, rhinitis, diverticulosis of large intestine without hemorrhage.

Additional concerns reported on interview include longstanding reduced hearing and smell, headaches that occur twice weekly, and tingling and numbness in his right buttocks that radiates to his right calf.

Neurological history

The patient played high school and college football. He sustained regular blows to the head but was unable to estimate the frequency with which this occurred. He experienced 5 blows to the head with loss of consciousness for less than 5 minutes. After all events, he returned to play within the same game and did not receive medical attention.

Medication

Vortioxetine 20 mg tab daily, Alprazolam 0.25 mg tablet as needed for anxiety and Trazodone 50 mg for sleep induction. Medication was tolerated well with a good response.

Bio-psychosocial history

Development: He has a twin who was breached at birth. The patient was born first. He reported no other known pre- or perinatal insults. Developmental milestones were reached

Table 1. Case reports of colpocephaly or porencephaly diagnosed in adulthood.

Authors	Demographics	Initial presentation
Colpocephaly		
Wunderlich et al. (1996)	60 yo female	New onset partial complex seizures
Cheong et al. (2012)	67 yo female	Headache and dizziness, comorbid meningioma
Esenwa & Leaf (2013)	60 yo female	Headache after head trauma
Ciurea et al. (2013)	28 yo female	Longstanding, intermittent headaches and vertigo
Brescian et al. (2014)	88 yo male	Left hand apraxia-new onset
Nasrat et al. (2015)	66 yo female	One month of declining mental status (co-morbid) abscess
Bartolome et al. (2016)	67 yo female	Syncopal episode
	60 yo female	Confusion with fever – upper respiratory infection
Parker et al. (2019)	29 yo male	Intermittent headaches
Porencephaly		
Van der Knaap et al. (2006)	24 yo female	Scan since children with porencephaly, intracranial hemorrhages later in life, porencephaly and COL4A1 mutation
Yang et al. (2012)	68 yo female	Otorrhea
Douzenis et al. (2010)	25 yo female	Delusions, insomnia
Hussain et al. (2015)	26 yo female	Psychosis
Noyan et al. (2016)	43 yo female	Brief psychotic episode
Thafar et al. (2017)	23 yo male	Involuntary movements of an upper limb/Simple partial seizure
Oommen et al. (2017)	28 yo female	Seizures, new onset
Qureshi et al. (2018)	56 yo male	Seizures, new onset

within expected timeframes. He reported hearing issues in elementary school. The patient completed a bachelor's degree with no known early attention/learning concerns. He was described as disengaged during education, but his history was not concerning for an early learning disorder. He works as an operations manager. No performance concerns have been noted by supervisors. He is married with two teenage daughters.

Family history

Amyotrophic lateral sclerosis (ALS) in his father with symptom onset in his 80s. His mother had a history of significant alcohol abuse and a possible alcohol-related dementia with symptom onset at approximately age 73.

Results

Neurological exam

Slight spasticity in right extremities. Left eye exophthalmia, mild with the full range of eye movements. No rigidity, bradykinesia or tremor noticed on neurological evaluation. No frontal release signs or concerns for apraxia were present on neurological assessment. EMG was negative for ALS pattern.

Brain MRI

Marked enlargement of the occipital horns of the lateral ventricles. Large porencephalic cyst in the left frontal and parietal lobes, communicating with the left occipital horn (See Figures 1–3).

Behavioral presentation

Reduced speech prosody but otherwise normal fluency and no observed word-finding problems or paraphasias. Mood was described as occasionally depressed and affect was restricted. Patient's thought processes were somewhat tangential,

without overt thought content pathology. Patient did not report perception difficulties.

Neuropsychological assessment

Abnormal neuropsychological profile due to weaknesses in executive functions, processing speed, and aspects of language. Within executive functions, he demonstrated impairments in impulsivity, cognitive flexibility, and set-shifting. His language profile was notable for impaired category fluency and a weakness in naming (see Table 2 for details).

Discussion

This case is unique and adds to the literature by examining our understanding of dual developmental neuroanatomical anomalies incidentally found in adulthood. Many aspects of our patient's history and exam findings are consistent with findings in the literature, including motor difficulties (Blackman et al., 1991), reduced hearing (Ryzenman et al., 2007), and concern for ongoing psychiatric symptoms (Hussain et al., 2015; Noyan et al., 2016). The current case is rare due to the incidental diagnosis in adulthood without significant motor, sensory, neurological, or cognitive impairments in early life.

Colpocephaly can be associated with partial or complete agenesis of the corpus callosum, Chiari malformations, lissencephaly, and microcephaly (Ciurea et al., 2013). The abnormal ventricular enlargement in colpocephaly is believed to be secondary to the developmental arrest of white matter formation that occurs during fetal development. Various etiologies have been proposed, including chromosomal abnormalities, intrauterine infection, perinatal anoxic-ischemic encephalopathy, intrauterine growth retardation, and maternal toxin exposure (Brescian et al., 2014; Garg, 1982; Herskowitz et al., 1985; Landman et al., 1989; Noorani et al., 1988). Colpocephaly discovered in asymptomatic adults is exceedingly rare (see Table 1).

Similarly to colpocephaly, porencephaly is typically diagnosed in infancy due to associated intellectual disability, seizures, motor abnormalities, or visual abnormalities. Diagnosis in adulthood is

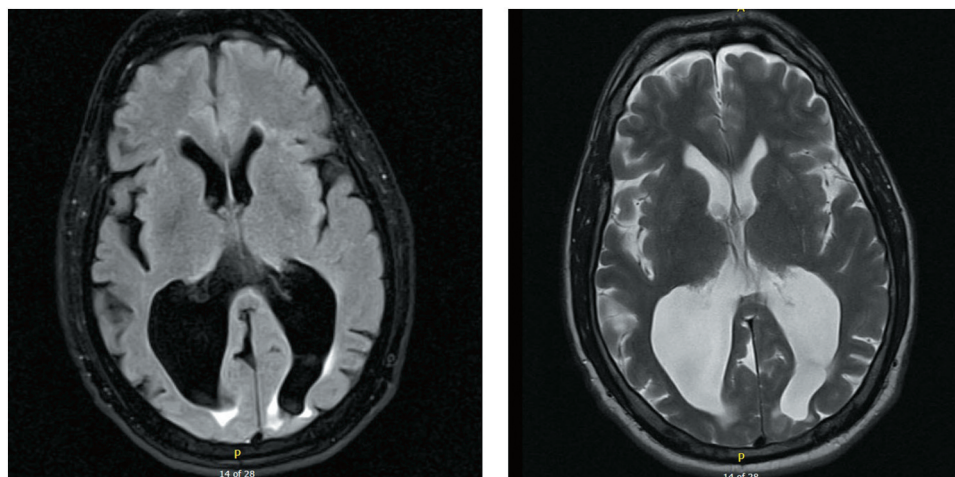


Figure 1. Brain MRI from the current case study: Axial FLAIR and corresponding T2 images showing enlarged occipital horns of lateral ventricles consistent with colpocephaly.

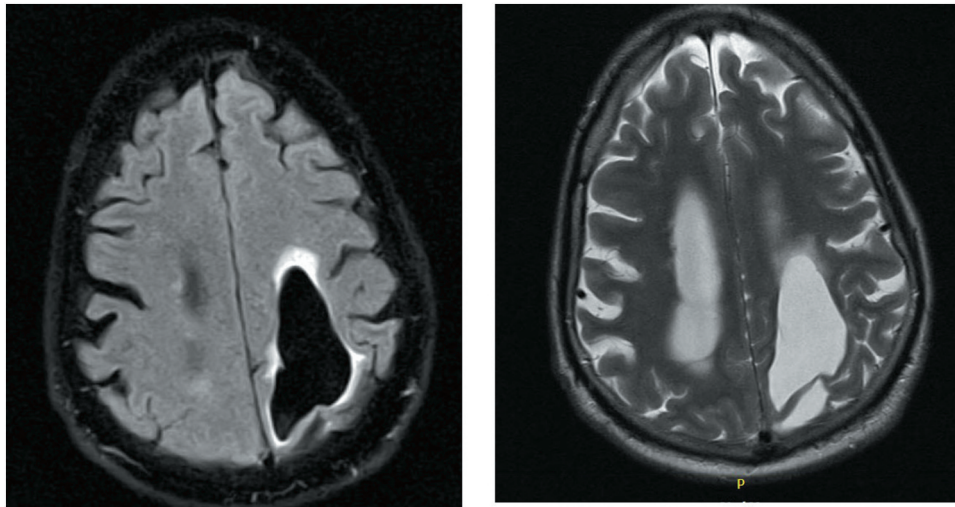


Figure 2. Brain MRI from the current case study: Axial FLAIR and corresponding T2 images showing large porencephalic cyst in left posterior frontal and parietal lobe.

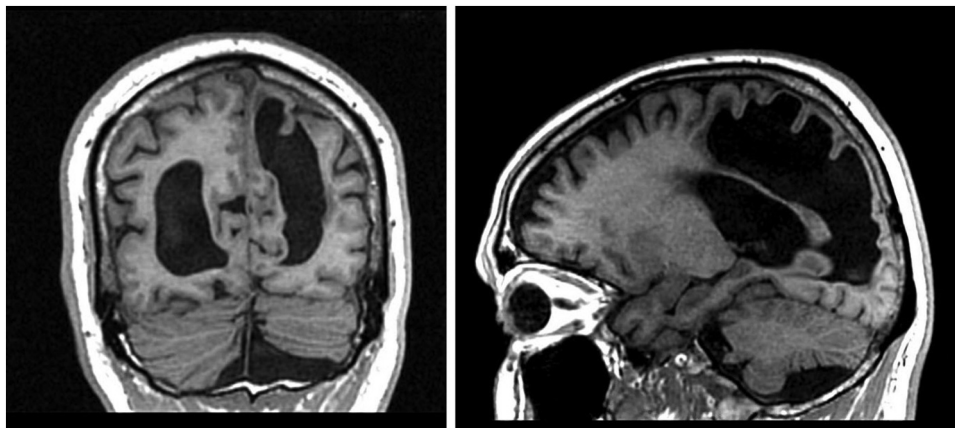


Figure 3. Brain MRI from the current case study: Coronal and sagittal FLAIR sequences showing large porencephalic cyst in the left frontal and parietal lobes, communicating with the left occipital horn.

remarkably uncommon and has only been reported only a few times previously (Bartolome et al., 2016; Brescian et al., 2014; Cheong et al., 2012; Esenwa & Leaf, 2013; Nasrat & Seraji-Bozoergzad, 2015; Wunderlich et al., 1996) (see Table 1).

The patient's neuropsychological evaluation revealed average IQ with weaknesses in executive functions, processing speed, and aspects of language. Given the paucity of literature related to cognitive functioning in the adult discovery of colpocephaly and porencephaly, it was unclear if the neuropsychological findings are typical of these diagnoses. Several studies revealed average IQ, consistent with our patient (Brescian et al., 2014; Danckert et al., 2004; Hussain et al., 2015), while other studies revealed below average IQ (Douzenis et al., 2010).

The patient's findings of executive dysfunction are consistent with his cyst locations in left frontal and parietal regions (Dixon et al., 2017; Scott & Schoenberg, 2011). Further, porencephaly in adults has been associated with executive dysfunction with cysts located in the left temporo-parietal-occipital region and left frontal-parietal regions (Chamard et al., 2016; Douzenis et al., 2010), somewhat consistent with our patient's MRI findings, though it should be noted that both studies findings were likely complicated by the presence of seizures

with the use of antiepileptic medications and psychosis, all of which can impact cognitive functioning.

Processing speed is generally thought to be associated with white matter integrity (Penke et al., 2010) and subcortical functioning (Andrade et al., 2012) which is not readily explained by the location of the patient's cysts; however, processing speed delays have been observed in groups of individuals with corpus callosum pathology (Marco et al., 2012). Interestingly, Marco et al. (2012) opined that impaired scores on executive functioning tasks could be explained by slowed processing speed, though the current patient had additional indicators of executive dysfunction, distinguishable from delayed processing speed, including errors within inhibition tasks and weaknesses on an untimed deductive reasoning task. Colpocephaly however is believed to be secondary to the developmental arrest of white matter formation during fetal development which may present clinically as slow speed of processing as well as executive dysfunction unrelated to the speed of processing (Girard et al., 1995).

The patient's language weaknesses were somewhat unexpected given that his language profile is indicative of left temporal dysfunction (Schoenberg & Scott, 2011; Troyer et al., 1998); however, the patient's brain MRI did not indicate

Table 2. Neuropsychological test results.

	Standard Score	Percentile	Label
<i>Wechsler Adult Intelligence Scale-IV</i>			
Prorated Full Scale IQ	92	30	Average
Verbal Comprehension Index	98	45	Average
Similarities	95	37	Average
Vocabulary	100	50	Average
Perceptual Reasoning Index	100	50	Average
Block Design	100	50	Average
Matrix Reasoning	100	50	Average
Working Memory Index	97	42	Average
Digit Span	105	63	Average
Arithmetic	90	25	Average
Processing Speed Index	76	5	Borderline
Symbol Search	80	9	Low Average
Coding	75	5	Borderline
<i>MEMORY</i>			
<i>Wechsler Memory Scale-IV</i>			
Logical Memory I	90	25	Average
Logical Memory II	90	25	Average
Logical Memory Recognition		>75	High Average
Visual Reproduction I	75	5	Borderline
Visual Reproduction II	85	16	Low Average
Visual Reproduction Recognition		17–25	Low Average
<i>Rey-Auditory Verbal Learning Task</i>			
Sum of Trials 1–5	79	8	Borderline
Immediate Recall	79	8	Borderline
Delayed Recall	77	6	Borderline
Recognition	109	73	Average
<i>Language</i>			
Boston Naming Test	78	7	Borderline
<i>Attention</i>			
<i>Conner's CPT-3</i>			
Detectability (d')	112		Slightly below average detect
Omissions	94		Average error rate
Commissions	122		Elevated error rate
Perseverations	104		Average
HRT	94		Average
HRT SD	97		Average consistency
Variability	96		Average variability
HRT Block Change	66		Stable reaction time throughout task
HRT ISI Change	91		Stable reaction times with changing task demands
<i>Executive Functions</i>			
Trail Making Test: Part-A	64	1	Impaired (1 error)
Trail Making Test: Part-B	79	8	Borderline (0 errors)
<i>Wisconsin Card Sorting Test</i>			
Categories Completed		6–10	Borderline
Failures to Maintain Set		<1	Impaired
Total Errors	80	9	Low Average
Perseverative Responses	92	30	Average
<i>Delis-Kaplan Executive Function System</i>			
<i>Verbal Fluency</i>			
Letter Fluency	85	16	Low Average
Category Fluency	55	0.3	Impaired
Switching Fluency	60	0.8	Impaired
Switching Accuracy	65	1	Impaired
<i>Color-Word Interference</i>			
Color Naming	65	1	Impaired
Word Reading	55	1	Impaired
Inhibition	85	16	Low Average
Inhibition/Switching	60	1	Impaired
Inhibition/Switching Errors	75	5	Borderline

anatomical abnormalities in this region. It is possible that the presence of left hemisphere porencephalic cysts places him at risk for language abnormalities. One case study in colpocephaly revealed a weakness in naming (Brescian et al., 2014), consistent with our findings; however, it should be noted that the patient was an octogenarian and thus, the presence of an underlying comorbid degenerative process can't be ruled out.

Interestingly, although the evaluations did not specifically assess for the presence of autism, there are some indications that corpus callosum pathology can be associated with

reduced social cognition (Lombardo et al., 2012), which was a concern reported by the patient. However, his behavioral presentation was inconsistent with ASD (i.e., no abnormalities in eye contact, reciprocity was observed in casual conversation), he met verbal milestones within expected timeframes, and he maintains current relationships.

A neurodegenerative process was considered for differential diagnosis. However, Alzheimer's disease, which is the most common neurodegenerative condition, was less favored due to grossly intact memory retrieval and storage on

neuropsychological evaluation, and lack of hippocampal/ mesial temporal atrophy on neuroimaging (Bondi et al., 2017). Additionally, he does not meet the criteria for Lewy body dementia or any other form of Parkinson's plus syndromes due to his neurological examination, and absence of cognitive fluctuations, well-formed visual hallucinations, and REM behavior disorder (McKeith et al., 2017). In terms of non-motor symptoms of Parkinsonism, the patient complained of anosmia which was however described as lifelong. We did not appreciate other symptoms of autonomic dysfunction (Goldstein, 2011). Frontotemporal dementia was ruled out based on behavioral observations and patient history that revealed no inappropriate behaviors, social disinhibition, stereotyped behaviors, or hyperorality (Rascovsky et al., 2011). Finally, Chronic Traumatic Encephalopathy (CTE) cannot be definitively ruled out in light of his history of multiple sports-related concussions; however, this possibility is also less favored. In particular, his described head injuries were primarily mild and did not require medical care. Further, he did not evidence motor changes most frequently attributed to CTE as described in the literature, including no significant bradykinesia, observed tremor, dysarthria, gaze disturbance, or ataxia (Reams et al., 2016).

Future directions for this patient include genetics testing, medication management, and repeat neuropsychological evaluation to assess his cognitive trajectory and aid in differential diagnosis. Future research may examine the late onset of motor changes, neurodevelopmental anomalies in the context of aging and cognitive decline, and assessing the susceptibility and cognitive compromise from additional neurological insults, such as head injury.

Disclosure statement

No potential conflict of interest was reported by the authors.

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